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| United States Patent | 12413147                   |
| Kind Code            | B2                         |
| Date of Patent       | September 09, 2025         |
| Inventor(s)          | Perreault; David J. et al. |

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### Variable controller and associated control methods

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#### Abstract

A circuit comprising a first capacitor configured to be charged to a voltage representing state information of a compensator, a second capacitor, a buffer circuit configured to charge the second capacitor to a voltage substantially equal to that of the first capacitor and a switching network configured to transition between a first state and a second state. When the switching network is in the first state, the second capacitor is charged to the voltage across the first capacitor. When the switching network is in the second state, the buffer circuit is disconnected from the second capacitor and the first capacitor and the second capacitor are connected in parallel.

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**Appl. No.:** 17/821257

**Filed:** August 22, 2022

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20230056740 A1          | Feb. 23, 2023           |

#### Related U.S. Application Data

us-provisional-application US 63235927 20210823

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## Publication Classification

**Int. Cl.:** H02M1/00 (20070101); H02M3/158 (20060101)

**U.S. Cl.:**

**CPC** H02M3/158 (20130101); H02M1/0012 (20210501); H02M1/0025 (20210501);

## Field of Classification Search

**CPC:** H02M (3/156); H02M (3/158); H02M (3/1582); H02M (3/335); H02M (3/33507); H02M (3/33569); H02M (3/33571); H02M (1/0003); H02M (1/0012); H02M (1/0016); H02M (1/0025); H02M (3/07); H02M (3/075)

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## **Background/Summary**

RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Application No. 63/235,927, filed Aug. 23, 2021, and entitled “VARIABLE CONTROLLER AND ASSOCIATED CONTROL METHODS,” which is incorporated herein by reference in its entirety.

### **FIELD**

(1) This disclosure relates generally to a variable controller and associated control methods, and more particularly to a variable controller and associated control methods for a compensator.

### **BACKGROUND**

(2) As is known in the art, adjusting a control system to improve its performance might lead to unexpected behavior. For example, increasing a gain value of a control system may result in poor stability (or even instability) of the system being controlled. To make the control system behave as desired, it may be necessary to include a compensator network (also sometimes referred to as a “compensator system,” a “compensator circuit,” or more simply a “compensator”) which compensates for deficient performance of an original control system. A compensator may thus be considered as component of a control system used to regulate other control system components. A compensator may perform such regulation, for example, by conditioning an input signal or an output signal of the system (e.g., conditioning an input or output signal to a control system).

(3) As is also known, a power electronic converter is a circuit which converts one form or level of electrical power to another form or level (e.g., a dc-to-dc converter). Many power electronic converters must operate over both a range of operating characteristics as well as in multiple, different operating modes. For example, many power electronic converters must operate over a range of input and/or output signals, voltage conversion ratios, ranging load characteristics, a range of power levels and the like as well as in multiple, different operating modes (e.g., buck, buck+buck-boost, buck/boost, boost, peak current mode, valley current mode, low-power modes such as pulse frequency modulation, etc.).

(4) The ability of a power converter to function over a range of operating characteristics and in multiple, different operating modes is often accomplished with a fixed controller or fixed compensator (i.e., a controller having a predefined architecture and controller structure such as a single-loop SISO control architecture where the controller is a fixed-order transfer function, a PID controller, or a PID controller plus a filter). However, because the dynamics of devices such as power converters vary considerably with operating point and operational mode, the achievable dynamic performance within a mode and/or over a range of power converter operating

characteristics can be quite limited by use of a single fixed controller or compensator.

## SUMMARY

(5) In accordance with the concepts described herein, it has been recognized that it would be valuable to provide a compensator that yields different dynamics/control performance for one or more of: (a) different operating modes (e.g., buck vs. boost vs. buck+buck-boost, or Pulse-Width Modulation vs. Pulse Frequency Modulation, etc.); and (b) different operating ranges or points (e.g., different input voltage, output voltage, reference voltage, conversion ratio or load regimes) such as different operating modes and/or different operating ranges or points of a power electronics circuit.

(6) In accordance with the concepts described herein, it has been recognized that it would be further desirable to be able to change compensator characteristics and/or dynamics such as pole and/or zero locations (i.e., the frequencies for which the value of the denominator and numerator of a transfer function becomes zero, respectively and which determine the stability of the system represented by the transfer function) without introducing transients or “bumps” in operation at the transition point when compensator characteristics are changed.

(7) Described is such a time/mode/operating point variable controller that is particularly well suited to integrated high-performance power electronic converter applications. It will also be appreciated that many other applications need operate across wide ranges and multiple modes and can likewise benefit from the techniques described herein.

(8) In an aspect, a circuit includes a first capacitor configured to be charged to a voltage representing state information of a compensator, a second capacitor, a buffer circuit configured to charge the second capacitor to a voltage substantially equal to that of the first capacitor, and a switching network configured to transition between a first state and a second state. When the switching network is in the first state, the second capacitor is charged to the voltage across the first capacitor. When the switching network is in the second state, the buffer circuit is disconnected from the second capacitor and the first capacitor and the second capacitor are connected in parallel.

(9) Implementations can include one or more of the following features.

(10) In some implementations, when the switching network is in the first state, dynamics of the compensator are determined at least in part by the first capacitor, and when the switching network is in the second state, the dynamics of the compensator are determined at least in part by a total capacitance provided by the first capacitor and the second capacitor connected in parallel.

(11) In some implementations, the capacitance provided based on the state of the switching network affects a pole location of the compensator.

(12) In some implementations, the capacitance provided based on the state of the switching network affects a zero location of the compensator.

(13) In some implementations, transitioning the switching network between the first and second states does not cause a transient impact on the voltage across the first capacitor.

(14) In some implementations, transitioning the switching network between the first and second states does not cause a transient impact on a voltage provided at an output of the compensator.

(15) In some implementations, transitioning the switching network between the first and second states does not disturb the voltage provided at the output of the compensator.

(16) In some implementations, the switching network comprises a complementary switch pair configured to switch between the first state and the second state.

(17) In some implementations, the circuit includes a third capacitor. The buffer circuit is configured to charge the third capacitor to a voltage substantially equal to the voltage across the first capacitor. The switching network is configured to transition between at least two additional states to cause the third capacitor to switch between being charged to the voltage across the first capacitor, and being disconnected from the buffer circuit and connected in parallel with one or both of the first capacitor and the second capacitor.

(18) In some implementations, the circuit includes a resistor connected to one or both of the first

capacitor and the second capacitor.

(19) In some implementations, the circuit is part of a power converter.

(20) In some implementations, when the complementary switch pair is in the first state, a first switch is closed and a second switch is open. When the complementary switch pair is in the second state, the first switch is open and the second switch is closed.

(21) In some implementations, the switching network is controlled by a control signal.

(22) In some implementations, the control signal automatically causes switches of the switching network to switch based on one or both of an operating mode and an operating point of the circuit.

(23) In some implementations, the operating mode includes buck, boost, or buck+buck-boost mode.

(24) In some implementations, the operating point includes one or more of an input voltage, an output voltage, a conversion ratio, and a load current of the circuit.

(25) In some implementations, the buffer circuit includes an op-amp.

(26) In some implementations, the op-amp is configured to charge the second capacitor to a voltage that is related to an output voltage and a reference voltage of the compensator.

(27) In an aspect, a compensator circuit is configured to provide an output voltage. The compensator circuit includes a plurality of resistors and a switching network coupled to at least one of the plurality of resistors. The switching network is configured to transition between states to cause selected ones of the plurality of resistors to be connected in parallel with each other. The compensator circuit also includes a capacitor coupled to the selected ones of the plurality of resistors connected in parallel with each other.

(28) Implementations can include one or more of the following features.

(29) In some implementations, dynamics of the compensator circuit are determined at least in part by a total resistance provided by the selected ones of the plurality of resistors connected in parallel with each other.

(30) In some implementations, the total resistance determined by the state of the switching network affects a zero location of the compensator circuit.

(31) In some implementations, the compensator circuit is part of a power converter.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The manner and process of making and using the disclosed embodiments may be appreciated by reference to the figures of the accompanying drawings. It should be appreciated that the components and structures illustrated in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principals of the concepts described herein. Like reference numerals designate corresponding parts throughout the different views. Furthermore, embodiments are illustrated by way of example and not limitation in the figures, in which:

(2) FIG. 1 is a block diagram of an exemplary control system for a power converter;

(3) FIG. 2 is a schematic diagram of an exemplary prior art compensator.

(4) FIG. 3 is a schematic diagram of an exemplary compensator provided in accordance with the concepts described herein in which a resistance value of a resistor  $R_{sub.c}$  can be dynamically changed;

(5) FIG. 4 is a schematic diagram of an exemplary compensator having a capacitor  $C_{sub.gm}$  which can be dynamically changed between a first capacitance value and a second capacitance value;

(6) FIG. 5 is a schematic diagram of an exemplary compensator having a capacitor  $C_{sub.gm}$  which can be dynamically changed among a plurality of capacitance values without disturbing a signal provided at an output of the compensator;

(7) FIG. 6A is a schematic diagram of an exemplary compensator that includes a plurality of

capacitive buffer circuits;

(8) FIG. 6B is a schematic diagram of an exemplary compensator in which the values of capacitive buffer circuits can each be dynamically changed (or reprogrammed) among two (2) values; and

(9) FIG. 7 is a schematic diagram of an exemplary compensator in which a capacitive value of a feedback capacitor C can be dynamically reprogrammed among two (2) values.

#### DETAILED DESCRIPTION

(10) Referring now to FIG. 1, an exemplary system **100** includes a control system **101** (sometimes referred to herein as a controller) for a power stage **116** (e.g., a dc-dc power converter). The system **100** may be a power electronics system. The control system **101** includes a selectable voltage divider **104** having an input coupled to an input **102** of control system **101**. A bandgap reference voltage  $V_{sub.egref}$  is provided to control system input **102** and coupled to an input of the selectable voltage divider **104**. Selectable voltage divider **104** receives the bandgap reference voltage  $V_{sub.egref}$  and provides a target reference voltage  $V_{sub.REF}$  at an output **106** thereof. As will become apparent from the description provided herein, target reference voltage  $V_{sub.REF}$  for an output voltage  $V_{sub.LN}$  **122** is set by bandgap voltage reference  $V_{sub.egref}$ . In embodiments, bandgap voltage reference  $V_{sub.egref}$  may have a value of about 1.25 V, which may be divided down by a programmable amount via selectable voltage divider **104**.

(11) An output of selectable voltage divider **104** is coupled to a first input of a difference circuit **107** and a second input of difference circuit **107** is coupled to receive a feedback voltage  $V_{sub.fb}$  **126**. An output of difference circuit **107** represents an error voltage  $V_{sub.err}$  **128** between the output voltage  $V_{sub.LN}$  **122** and a desired value of the output voltage. The output of difference circuit **107** is coupled to an input of a compensator **108** (sometimes referred to as an error amplifier), and an output of compensator **108** is coupled to a current control circuit **112**. An output of current control circuit **112** is coupled to the power stage **116** and power stage **116** provides an output voltage  $V_{sub.LN}$  at an output **122** thereof. Power stage **116** may be provided, for example, as a power converter. It should, of course, be appreciated that although the broad concepts described herein are described in the context of a power electronics system, control systems (such as control system **101**) operating in accordance with the concepts described herein may be used in conjunction with other types of systems (i.e., systems other than power electronics systems). Such other systems can include servomechanisms (e.g., position or velocity control systems), tracking or targeting systems, and actuation systems, among many others.

(12) In some implementations, the compensator **108** may be referred to as an error amplifier herein. In some implementations, the components of the control system **101** may be referred to cumulatively as a compensator, and an error amplifier can be a part of the overall compensator.

(13) At least a portion of output voltage  $V_{sub.LN}$  **122** is coupled to an input of a voltage divider circuit **124** and a feedback voltage  $V_{sub.fb}$  **126** is provided at an output of voltage divider circuit **124**. Thus, feedback voltage  $V_{sub.fb}$  **126** (which may be a divided down version of the output voltage  $V_{sub.LN}$  **122** provided by voltage divider **124**) is provided to a second input of difference circuit **107** and compared with reference voltage  $V_{sub.REF}$  to provide an error voltage  $V_{sub.err}$  **128**.

(14) Error voltage  $V_{sub.err}$  **128** is coupled to an input of compensator **108**. Compensator **108** provides an output signal (here, an output voltage  $V_{sub.ea\_comp}$ ) at an output **110** thereof. Compensator output signal is coupled through current control circuit **112** to an input of power stage **116** and is used to control power stage **116**. Such control system output signals may correspond, for example, to signals for commanding a duty ratio in a duty ratio-controlled converter (also known as voltage-mode control), commanding a peak commanded current for the inductor in a peak current-mode-controlled converter, or another control variable (e.g., duty ratio, valley current, average current, etc.).

(15) In embodiments, an associated modulator (duty cycle generator, PWM unit, current-mode control block, etc.) may use such control signals (e.g., commands) to generate switching signals for

the power stage **116**, as is known in the art.

(16) An output signal from power stage **116** is coupled through a scaling and offset circuit **118** to an input of current control circuit **112** as a current signal  $i_{\text{sub.sense}}$ . A ramp generator circuit **120** also provides a signal to current control circuit **112**. It should be appreciated that circuits **118**, **120** are auxiliary to the controller and thus in some embodiments may not be included or may not be considered part of control system **101**.

(17) The current control circuit **112** provides a way to regulate inductor current in a power converter. The current control circuit **112** takes as inputs a desired reference for said inductor current  $V_{\text{sub.ea\_comp}}$ , provided at the output **110** of the compensator **108**, and a measurement or estimate of an inductor or other current ( $i_{\text{sub.sense}}$ ) in the power converter power stage **116**, generates as an output a switching signal  $q(t)$  for said power converter power stage **116**. The current control circuit **112** can be useful for providing multi-loop control or full-state feedback control of a power converter for improved dynamic performance. Scaling and offset circuit **118** allows a measured or estimated value of the inductor or other current ( $\lambda_{\text{sub.L}}$ ) in power stage **116** to be converted to a form and range that is useful for the current control circuit **112** (i.e.,  $i_{\text{sub.sense}}$ ). Ramp generator **120** can provide a timing (clocking) reference and ramp to the current control circuit **112** allow the current control circuit **112** to implement slope compensation.

(18) Characteristics of the compensator **108** can be selected based upon the dynamics of the “plant.” In some implementations, the plant comprises at least the power stage **116**. The characteristics can be selected to result in stable control of the output of the system **100**, preferably with fast transient response and good stability margin. The plant dynamics (i.e., the open-loop dynamics of power stage **116**), and hence the overall system dynamics (i.e., the closed loop system dynamics of the plant—i.e., power stage **116**—modified by the compensator **108**), can depend upon converter operating mode (buck, boost, PFM, PWM, etc.) as well as operating point (output voltage, load, conversion ratio, etc.) making the design of the compensator **108** challenging.

(19) The compensator **108** of FIG. 1 may take the form of or may include one or more features of any one or more of the compensators described herein with respect to FIGS. 3-7.

(20) Referring now to FIG. 2, FIG. 2 shows a circuit diagram of a prior art compensator **200**. Such a circuit might be implemented as an integrated circuit (IC) such as part of a power converter IC or as a controller IC.

(21) Compensator **200** comprise an amplifier **202** which is here represented as a model having a gain element **202a** and an output resistance  $R_{\text{sub.o}}$  **202b**. In embodiments, amplifier **202** may, for example, be provided as a transconductance amplifier (e.g., an operational transconductance amplifier) or a voltage amplifier. A plurality of resistive and/or capacitive signal paths (or “legs”) **205**, **209** are coupled at an output of amplifier **202**, for example, in the form of an output voltage  $V_{\text{sub.ea\_comp}}$ . In this example, signal path **205** comprises series coupled resistor  $R_{\text{sub.c}}$  **206** and a capacitor  $C_{\text{sub.c}}$  **208** and signal path **209** comprises a capacitor  $C_{\text{sub.gm}}$  **210**. In embodiments, compensator **200** may be provided as an integrated circuit.

(22) In some implementations, amplifier **202** might be realized as an operational amplifier. In some implementations, the value of gain (“transconductance”)  $g_{\text{sub.m}}$  might be made programmable over a range in order to allow some degree of programmability/dynamic tuning of the system dynamics.

(23) An error voltage  $V_{\text{sub.err}}$  **228** is provided to an input of compensator **200** and coupled to amplifier **202**.

(24) If one assumed that amplifier output resistance  $R_{\text{sub.o}}$  **202b** were infinite, a small-signal compensator transfer function  $G_{\text{sub.C}}(s)$  from  $V_{\text{sub.err}} = V_{\text{sub.REF}} - V_{\text{sub.fb}}$  to the compensator output **110**  $v_{\text{sub.ea\_comp}}$  could be expressed as:

(25) 
$$G_C(s) = \frac{V_{\text{ea\_comp}}}{V_{\text{err}}} = \frac{g_m}{s(C_{gm} + C_c)} \cdot \frac{1 + sR_c C_c}{1 + sR_c \frac{C_{gm} C_c}{C_{gm} + C_c}} \quad (\text{Eq. 1})$$

(26) In practical circuits, however, the output resistance  $R_{\text{sub.o}}$  **202b** is finite. However, if the

amplifier output resistance is sufficiently large (e.g., such that the error between (Eq. 1) and (Eq. 2) is unimportant), this expression is useful for approximately locating the compensator zero and second pole. Including output resistor  $R_{o202b}$ , the following compensator transfer function results:

$$(27) \quad G_C(s) = \frac{v_{ea\_comp}}{v_{err}} = g_m R_o \cdot \frac{1 + sR_c C_c}{s^2 R_o C_{gm} R_c C_c + s(R_o C_{gm} + R_o C_c + R_c C_c) + 1} \quad (\text{Eq. 2})$$

(28) In this exemplary compensator **200**, the compensator DC gain is set by the product of amplifier gain  $g_{sub.m}$  and output resistance (i.e., the compensator DC gain is set by the product of  $g_{sub.m} R_{sub.o}$ ). The value of resistor  $R_{sub.c206}$  and capacitor  $C_{sub.c208}$  set the zero of the transfer function, while the pole locations are set by multiple component parameters as shown in Eq. 2. One can adjust compensator dynamics (i.e., the manner in which the compensator responds—in an electrical circuit sense—to inputs provided thereto) by introducing programmability of one or more of these elements (e.g., by using a switchable resistor bank to dynamically adjust (or change) the resistance value of resistor  $R_{sub.c206}$  to modify the compensator zero, by dynamically programming the gain value  $g_{sub.m}$  to modify the compensator gain, etc.). Such dynamic adjustments/changes/programming can be based on system operating condition (e.g., mode, operating voltages or power levels, etc.). It can be desirable to have different values of compensator components to modify how the circuit will react dynamically under different system operating conditions, and such dynamic adjustability can allow such.

(29) The capacitor  $C_{sub.gm210}$  can be a combination of device output capacitance and physical capacitance used in setting characteristics of the transfer function  $G_{sub.C}(s)$  as per (Eq. 2). The resistor  $R_{sub.c206}$  and the capacitor  $C_{sub.c208}$  are circuit elements used to set the transfer function as per (Eq. 2). Using the components as illustrated allows the ability to set different characteristics of the transfer function  $G_{sub.C}(s)$ .

(30) Referring now to FIG. 3, in this embodiment, an exemplary compensator **300** includes a programmable resistive element **306** (i.e., a resistive element capable of being adjusted or changed) comprising a plurality (or bank) of parallel coupled resistive legs **305a-405N**. At least one of the resistive legs (illustrated as a single resistive leg **305a** in FIG. 3) comprises a resistor (here illustrated as resistor **306a**) having a resistance value  $R_{sub.c0}$ . The compensator **300** also includes a switching network that can include a plurality of switch elements, or switches  $S_{sub.1}$ - $S_{sub.N}$  **307b-307N**.

(31) Programmable resistive element **306** further comprises at least one resistive leg comprising both a resistive element and a switch element. In the example embodiment of FIG. 3, resistive legs **305b-305N** each comprise both a resistive element and a switching element. Taking resistive leg **306b** as illustrative of resistive legs **305c-305N**, resistive leg **305b** comprises a resistor **306b** having a resistance value  $R_{sub.c1}$  coupled to a switch  $S_{sub.1}$  **307b**. Similarly, resistive legs **305c-305N** comprise respective ones of resistors **306c-306N** having respective resistance values  $R_{sub.c2}$ - $R_{sub.cN}$  coupled to respective ones of switches  $S_{sub.2}$ - $S_{sub.N}$  **307c-307N**.

(32) When switches  $S_{sub.1}, \dots, S_{sub.N}$  **307b-N** are closed or on (i.e., the switch is biased or otherwise set into a state or position such that a low impedance signal path (and ideally a short circuit impedance signal path) exists between a pair of terminals thereof) a conductive path having a resistance less than an open circuit resistance exists between nodes **313a, 313b**. For example, with switch  $S_{sub.1}$  closed or on and all other switches  $S_{sub.2} \dots S_{sub.N}$  open or off, a signal path comprising resistor **306b** and switch **307b** having a resistance substantially equal to  $R_{sub.c1}$  exists between nodes **313a, 313b**. Thus, by selecting which of switches  $S_{sub.1}, \dots, S_{sub.N}$  **307b-N** are turned on, resistive element **306** is adjustable (or programmable or selectable) in steps between a first (or minimum) resistance value and a second (or maximum) resistance value.

(33) In the example embodiment of FIG. 3, the maximum resistance value is set by the resistance value  $R_{sub.c0}$  of resistor **306a** and the minimum resistance value is set by the equivalent resistance provided by resistors **306a-306N** (i.e., the equivalent resistance provided by the parallel

combination of resistors R.sub.c0||R.sub.c1||. . . ||R.sub.cn). Thus, by selecting which of switches S.sub.1, . . . , S.sub.N **307b-N** are turned on (i.e., biased or otherwise set into their ON states) resistive element **306** is provided having one of a plurality of different resistance characteristics. (34) The plurality of switches S.sub.1-S.sub.N **307b-307N** can thus operate (or be controlled) to cause one or more of the plurality of second resistors R.sub.c1-R.sub.cn **306b-306N** to either be electrically disconnected from the first resistor R.sub.c0 **306a** or coupled in parallel with the first resistor R.sub.c0 **306a**. A total resistance provided by the first resistor R.sub.c0 **306a** and coupled ones of the plurality of second resistors R.sub.c1-R.sub.cn **306b-306N** can cause one or more dynamics of the compensator **300** to be adjusted, such as one or more zero locations of the compensator **300**, as described herein.

(35) As noted above, resistive element **306** has a first terminal coupled to a node **313a** at which a compensator output **110** (e.g., output voltage V.sub.ea\_comp) is provided. For example, the voltage at node **313a** with reference to ground may correspond to the compensator output **110** (e.g., the output voltage V.sub.ea\_comp). A second terminal of resistor **306a** is coupled to a node **313b**. A first terminal of a capacitor **308** having a capacitance of C.sub.c is coupled to a node **313b** and a second terminal of capacitor **308** coupled to a reference potential, here ground.

(36) It should, of course, be appreciated that while FIG. 3 illustrates one example embodiment of a dynamically programmable element (i.e., a dynamically programmable resistance element **306**) disposed to influence (e.g., change or shape) the dynamics of a compensator, other embodiments which implement the concept of dynamic programmability are also possible. For example, an implementation comprising a programmable resistance element may comprise a first signal path having a set of individual resistors coupled in series, a second signal path coupled in parallel to the first signal path and or more control switches (e.g., one or more of switches **307b-307N**) disposed at various points between the first and second signal paths and operable to selectively include or bypass one, some or all of the resistors in the first signal such that only selected ones of the resistors in the first signal path contribute to the overall resistance between first and second terminals of the programmable resistance element and thus can be used to adjust one or more dynamics of the compensator. For example, the one or more dynamics that can be adjusted can include one or more zero locations of the compensator.

(37) In another example embodiment, the plurality of second resistors R.sub.c1-N **306b-N** and the plurality of switches S.sub.1-N **307b-N** can be arranged such that the switches **307b-N** operate to couple selected ones of resistors R.sub.c1-N **306b-N** in series with the first resistor R.sub.c0 **306a**. A total resistance provided by the first resistor R.sub.c0 **306a** and the plurality of second resistors R.sub.c1-N **306b-N** can cause one or more dynamics of the compensator **300** to be adjusted. For example, the one or more dynamics that can be adjusted can include one or more zero locations.

(38) The above-described programmability can be such that a dynamic (transient) impact on the system does not result during operation. In the embodiment of FIG. 3, compensator **300** comprises a programmable resistance element and the compensator **300** state is held by the voltage on capacitors C.sub.gm **311** and C.sub.c **308**, with capacitor C.sub.gm **311** being particularly central to determining the compensator pole locations and resulting system dynamics. Stated differently, in the system of FIG. 3, the compensator output **110** V.sub.ea\_comp is substantially, directly determined by the voltage on capacitor C.sub.gm **311**. In this way, the voltage that the capacitor C.sub.gm **311** is charged to can represent state information (e.g., the compensator dynamics described herein such as gain, pole locations, zero locations, etc.) for the compensator **300**. Any operation (e.g., compensator programming change) that disturbs this voltage causes a variation (e.g., a “bump”) in the compensator output **110** V.sub.ea\_comp. Such variations in compensator output signals (e.g., output voltage V.sub.ea\_comp) are undesirable for control.

(39) Thus, in accordance with a further aspect of the concepts described herein, it has been recognized that it would be desirable to provide a system that allows dynamic reprogrammability/variability of compensator capacitors (such as capacitor C.sub.gm **311**) to

enable rapid modification of control (and associated dynamics) for different modes and operating ranges without introducing control bumps in the compensator output voltage  $V_{\text{sub.ea\_comp}}$ . (40) In accordance with a still further aspect of the concepts described herein, it has also been recognized that the above goal may be achieved by introducing one or more additional compensation capacitors that can be switched in and out (i.e., electrically coupled to and electrically decoupled from) a compensator network thus making the compensation capacitor dynamically programmable.

(41) As will be described in detail below, to prevent undesirable transients when a compensation capacitor is switched into the compensator, a buffer may be utilized that keeps compensation capacitors that are not presently active in the compensator pre-charged at the value of the compensator capacitor voltage in use (i.e., tracking the active compensator capacitor's voltage value.) An exemplary circuit for realizing a programmable value for a capacitor (e.g., such as capacitor **311** in FIG. 3 having capacitance value  $C_{\text{sub.gm}}$ ) is shown in FIG. 4.

(42) Referring now to FIG. 4, shown is an exemplary compensator **400** having an effective capacitance which can be dynamically programmed (i.e., changed or otherwise adjusted during operation) between at least a first capacitance value and a second, different capacitance value. In the example of FIG. 4, the capacitance of compensator **400** can be dynamically programmed between a first capacitance value corresponding to  $C_{\text{sub.gm1}}$  and a second capacitance value corresponding to  $C_{\text{sub.gm1}} + C_{\text{sub.gm2}}$ . Significantly, the effective capacitance value of compensator **400** (generally referred to as  $C_{\text{sub.gm}}$ ) can be programmed without disturbing the compensator output signal **110** (e.g., without causing a transient impact on the compensator output voltage  $V_{\text{sub.ea\_comp}}$ ).

(43) In this example embodiment, compensator **400** includes a first capacitor **411a** and a capacitive buffer circuit **415**. In this example, the capacitive buffer circuit **415** includes a second capacitor **411b**. First capacitor **411a** is provided having a capacitance corresponding to  $C_{\text{sub.gm1}}$  and the second capacitor **411b** is provided having a capacitance corresponding to  $C_{\text{sub.gm2}}$ . Capacitive buffer circuit **415** also comprises a switching network **416** coupled to the second capacitor **411b**. Switching network **416** has two states: a first state for charging the second capacitor **411b** to a desired value, and another state for placing capacitor **411b** in parallel with capacitor **411a**. It should be appreciated that the functionality of switching network **416** may be achieved with a variety of different circuit implementations. For example, as illustrated in the example embodiment of FIG. 4, such a switching network **416** may be implemented via a single complementary switch pair such as with switches **431a**, **431b**. In FIG. 4, switches **431a**, **431b** are configured to switch in a complementary fashion (i.e., when switch **431a** is biased into its ON state, switch **431b** is biased into its OFF state and vice-versa) and thus switches **431a**, **431b** are sometimes referred to as a “complementary switch pair” or as “complementary switches.” After reading the disclosure provided herein, those of ordinary skill in the art will appreciate how best to implement the functionality provided by a switching network for use in a particular application.

(44) Complementary switch pair **431a**, **431b** are controlled by a switching signal  $q(t)$  (also sometimes referred to as a “switch control signal,” a “programming signal” or more simply a “control signal”). The complementary switch pair **431a**, **431b** are coupled to a first capacitor **411a** having a capacitance of  $C_{\text{sub.gm1}}$ .

(45) In one example embodiment, buffer circuit **415** is configured to receive a switch control signal  $q(t)$  such that when switch control signal  $q(t)$  has a first value (e.g.,  $q(t)$  is low such as a logical 0), switch **431a** is closed (or placed in its ON state) and switch **431b** is open (or placed in its OFF state).

(46) In this circuit configuration (i.e., control signal  $q(t)$  has the first value thereby placing switch **431a** in closed or ON state and switch **431b** in an open or OFF state), capacitor **411a** is coupled to a buffer **412** that ensures its voltage tracks that of the output voltage **110**, and includes switch circuitry which effectively electrically disconnects (or electrically removes) capacitor **411a** from

the compensator circuit **400** (e.g., such that the capacitor **411a** having capacitance value  $C_{sub.gm1}$  does not influence the dynamics of the compensator **400**). In embodiments, the buffer **412** may comprise an op-amp follower circuit or more simply “a follower” that maintains the voltage on the first capacitor **411a** to be substantially the same as the voltage on a second capacitor **411b**, which is also substantially the compensator output voltage  $V_{sub.ea\_comp}$  **110**.

(47) A first input of buffer **412** (here a positive input) can be coupled to a node **413** (e.g., a positive node of the compensator output **110** with reference to ground) and be configured to charge the first capacitor **411a** to substantially the same voltage provided at the node **413** (e.g., the compensator output voltage  $V_{sub.ea\_comp}$ ).

(48) Continuing with the above example, when control signal  $q(t)$  has a second value different from the first value (e.g., control signal  $q(t)$  is high or a logical 1), the first switch **431a** is open (or biased or otherwise set into its open or off state) and the second switch **431b** is closed (or biased or otherwise set into its closed or on state) such that the first capacitor **411a** is electrically disconnected from the buffer **412** and is coupled in parallel with capacitor **411b** such that the compensator has an effective capacitance value of  $C_{sub.gm} = C_{sub.gm1} + C_{sub.gm2}$ .

(49) As noted above, the complementary switches **431a**, **431b** are configured to transition between a first state and a second state. When the complementary switches **431a**, **431b** are in the first state, the first capacitor  $C_{sub.gm1}$  **411a** can be charged by buffer **412** to the voltage provided at the node **413** (e.g., the compensator output voltage  $V_{sub.ea\_comp}$ ) without capacitor  $C_{sub.gm1}$  **411a** affecting how the compensator behaves. When the complementary switches **431a**, **431b** are in the second state, the first capacitor  $C_{sub.gm1}$  **411a** and the second capacitor  $C_{sub.gm2}$  **411b** are coupled in parallel to the node, operating together as a single capacitor.

(50) In embodiments, when the switch control signal  $q(t)$  changes from a first value to a second different value (e.g., from a logical 0 to a logical 1), the effective compensator capacitor  $C_{sub.gm}$  **411** changes to  $C_{sub.gm1} + C_{sub.gm2}$  from  $C_{sub.gm1}$ . However, because capacitor  $C_{sub.gm2}$  **411b** may be appropriately precharged by the buffer **412**, the compensator output voltage  $V_{sub.ea\_comp}$  is not disturbed (e.g., no transient impact is caused on the voltage across  $C_{sub.gm1}$  **411a** and/or  $C_{sub.gm2}$  **411b**). Likewise, when switch control signal  $q(t)$  changes from the second, different value to the first value (e.g., from a logical 1 to a logical 0), the effective compensator capacitor **411** changes to  $C_{sub.gm1}$  from  $C_{sub.gm1} + C_{sub.gm2}$  without disturbing the compensator output. Thus, depending on the state of the complementary switch pair and the total effective compensator capacitance provided by capacitor **411** determined by switch control signal  $q(t)$ , different pole locations for the compensator may be provided. In this way, the total effective compensator capacitance provided by capacitor **411** can cause one or more dynamics of the compensator **400** to be adjusted.

(51) It will be appreciated that with additional capacitors and complementary switch pairs (with corresponding control signals) the range of programmability of a compensation capacitor may be increased.

(52) FIG. 5, for example, shows a circuit diagram of an exemplary compensator **500** comprising a plurality of compensation capacitors. In this example embodiment, compensator **500** comprises three (3) compensation capacitors. After reading the disclosure provided herein, those of ordinary skill in the art will appreciate that compensator **500** may comprise fewer or more than three compensation capacitors. In this example, compensator **500** comprises a first compensation capacitor **511a** having a capacitance value of  $C_{sub.gm1}$ , a second compensation capacitor **511b** having a capacitance value of  $C_{sub.gm2}$ , and a third compensation capacitor **511c** having a capacitance value of  $C_{sub.gm3}$ .

(53) With the plurality of compensation capacitors in the example embodiment of FIG. 5, the effective capacitance value of compensator **500** can be dynamically reprogrammed (i.e. adjusted or changed) among 4 values:  $C_{sub.gm1}$ ,  $(C_{sub.gm1} + C_{sub.gm2})$ ,  $(C_{sub.gm1} + C_{sub.gm3})$ , and  $(C_{sub.gm1} + C_{sub.gm2} + C_{sub.gm3})$  without disturbing (i.e., without disrupting or otherwise

substantially changing) the voltage  $V_{sub.ea\_comp}$  at the compensator output **110**.

(54) In this example embodiment, programming (or adjustment or changing) of the compensator capacitance is done via two control signals  $q_{sub.2}(t)$  and  $q_{sub.3}(t)$  (i.e., switch control signals) that control a switching network that can include complementary switch pairs **531a, b** (generally denoted **531**), **533a, b** (generally denoted **533**). Each of the control signals  $q_{sub.2}(t)$  and  $q_{sub.3}(t)$  controls one of the complementary switch pairs **531, 533**, and the control signals  $q_{sub.2}(t)$  and  $q_{sub.3}(t)$  cause corresponding capacitors **511b** and **511c** to be switched between participating in the compensation and being precharged by a buffer **512**. The complementary switch pairs **531, 533**, the buffer **512**, and the capacitors  $C_{sub.gm2}$ - $C_{sub.gm3}$  **511b-c** are part of a capacitive buffer circuit **515** of the compensator **500**. In this example, control signal  $q_{sub.2}(t)$  controls complementary switch pair **531a, 531b** and control signal  $q_{sub.3}(t)$  controls complementary switch pair **533a, 533b**.

(55) The first complementary switch pair **531a, 531b** is coupled to the capacitor  $C_{sub.gm2}$  **511b** and second complementary switch pair **533a, 533b** is coupled to the capacitor  $C_{sub.gm3}$  **511c** (sometimes referred to herein as a third capacitor, e.g., relative to the first and second capacitors of FIG. 4). The first complementary switch pair **531a, 531b** and the capacitor **511b** are configured to operate similarly to the first capacitor **411a** and the complementary switch pair **431** of FIG. 4.

(56) The second complementary switch pair **533** is configured to transition between a first state and a second state. When the second complementary switch pair **533** is in the first state, a first switch **533a** is closed (i.e., switch **533a** is in its ON state) and a second switch **533b** is open (i.e., switch **533b** is in its OFF state) such that the third capacitor **511c** can be charged to a voltage provided at a node **513** to which a positive input of the buffer **512** is coupled. In the first state, the third capacitor  $C_{sub.gm3}$  **511c** can be coupled to an output of the buffer **512** to charge to the voltage  $V_{sub.ea\_comp}$  provided at the compensator output **110**. The compensator output voltage  $V_{sub.ea\_comp}$  is provided at the node **513** relative to ground. When the second complementary switch pair **533** is in the second state, the first switch **533a** is open and the second switch **533b** is closed such that the third capacitor  $C_{sub.gm3}$  **511c** can be coupled to the node **513** in parallel with  $C_{sub.gm2}$  **511b** and/or  $C_{sub.gm1}$  **511a** (e.g., which can correspond to the first and second capacitors of FIG. 4).

(57) Generally, augmenting a base capacitor (e.g., capacitor **511a**) with  $N$  complementary switch pairs and associated programming capacitors will allow selection among up to  $2^{sup.N}$  programmable values of compensation capacitance.

(58) It will be further appreciated that compensator designs can be implemented having multiple programmable compensation capacitors.

(59) Referring now to FIG. 6A in which like elements of FIG. 4 are provided having like reference designations, shown is a circuit diagram of an exemplary compensator **600** that may be generally similar to the compensators described in the preceding figures. The circuit diagram expresses the concept that the compensators described herein may include one or more capacitive buffer circuits **615a, 615b, . . . 615n**. Each of the capacitive buffer circuits **615** can include at least one or more capacitors, one or more resistors, one or more buffers, and a switching network that can include one or more switches that are configured to couple the elements in a way that causes one or more of the dynamics of the circuit to be adjusted without causing a transient impact on the output voltage  $V_{sub.ea\_comp}$  at the compensator output **110**.

(60) Referring now to FIG. 6B in which like elements of FIG. 6A are provided having like reference designations, a circuit diagram of an exemplary compensator **601** in which the values of compensator capacitors  $C_{sub.gm}$  **611** and  $C_{sub.c}$  **608** are each dynamically reprogrammable among two values. Compensator **601** comprises first and second capacitive buffer circuits **615a, 615b**. The right-half of the circuit diagram comprises second capacitive buffer circuit **615b** which may be the same as or similar to capacitive buffer circuit **615b** shown in FIG. 6A). Capacitive buffer circuit **615b** may operate in a manner which is the same as or similar to the capacitive buffer

circuit described above in conjunction with FIG. 4, with a buffer **612** configured to charge a capacitor **611b** and a switching network including a complementary switch pair **631** configured to switch between two states to cause the capacitor **611b** to switch between being charged by the buffer **612** and being coupled to another capacitor **611a** in parallel.

(61) The left-half of the circuit diagram, corresponding to capacitive buffer circuit **615a**, which may be the same as or similar to capacitive buffer circuit **615b** shown in FIG. 6A, employs a concept similar to the ones described with respect to the preceding figures. In particular, the compensator **600** includes a first capacitor C.sub.c1 **608a** that is coupled to a positive input of a buffer **607**, a second capacitor C.sub.c2 **608b** coupled to an output of the buffer **607**, and a complementary switch pair **609** configured to alternate between a first state and a second state. In the first state, a first switch **609a** is closed and a second switch **609b** is opened such that the second capacitor C.sub.c2 **608b** charges to a voltage equal to a voltage across the first capacitor C.sub.c1 **608a**. In the second state, the first switch **609a** is open and the second switch **609b** is closed such that the second capacitor C.sub.c2 **608b** is coupled in parallel with the first capacitor C.sub.c1 **608a**. When coupled in parallel, the combined capacitance of the first capacitor C.sub.c1 **608a** and the second capacitor C.sub.c2 **608b** is coupled in series with the resistor R.sub.c **406**.

(62) A control signal q.sub.A can be used for programming the capacitance used for capacitors C.sub.c1 **608a** and C.sub.c2 **608b**, and a control signal q.sub.B can be used for programming the capacitance used for capacitors C.sub.gm1 **611a** and C.sub.gm2 **611b**. One could likewise have circuits in which each programmable in multiple steps as illustrated in the preceding figures.

(63) The capacitor value(s) used in the compensator can be automatically adjusted based on operating mode (e.g., buck vs. boost, etc.), operating point (e.g., input voltage, output voltage, conversion ratio, load current value, etc.) or other range associated with converter operation, or based on associated reference or target values. Alternatively or additionally, the values may be made software or firmware programmable, e.g. using registers that control the control signals q(t).

(64) While the above examples focus on various types of compensators and implementations that are easily realized for IC power converter designs, it will be appreciated that a variety of compensator types and implementations can benefit from the programmability described herein. For example, one can realize a variety of compensator types with programmable capacitor values (e.g., integral, lead, lag, lead-lag, PI, PID, PD, type I, type II, type III, etc.) Moreover, other circuit implementations for compensators can be used, such as compensator designs using an op amp as an error amplifier.

(65) FIG. 7 shows a circuit diagram of an exemplary compensator **700** based on an op-amp **709** in which a net feedback capacitance C is provided by various combinations of capacitors **704a** **704b** (generally denoted **704**). The net capacitance value C can be dynamically selected to correspond to one of two values (C.sub.1 and C.sub.1+C.sub.2) via control signal q(t). The compensator **700** includes a switching network that can include a complementary switch pair **705a**, **705b** configured to switch between a first state and a second state. In the first state, first switch **705a** is closed and second switch **705b** is open such that capacitor C.sub.2 **704b** is charged to a reference voltage V.sub.REF **706** by a buffer **712**. In this way, capacitor C.sub.1 **704a** is pre-charged to the voltage it would have in the compensator **700** when it participates in the compensator **700** (e.g., when it is switched into the compensator **700** in such a way that it can have an influence on dynamics of the compensator **700**). This voltage can have the same value as the “active” capacitor it will be placed in parallel with (e.g., capacitor C.sub.2 **704b**). In some implementations, the capacitor C.sub.1 **704a** is pre-charged to a voltage related to the output voltage V.sub.ea\_comp **110** and the feedback voltage V.sub.fb **126**. In the second state, first switch **705a** is open and second switch **705b** is closed such that capacitor C.sub.1 **704a** is coupled in parallel with capacitor C.sub.2 **704b**. In this way, the dynamics of the compensator **700** can be adjusted without causing a transient impact on the compensator output voltage V.sub.ea\_comp **110**.

(66) It will be appreciated that while this is a type I or integral controller, other controller types

(including, but not limited to on/off controllers, proportional controllers derivative controllers) with one or more programmable circuit elements (including, but not limited to one or more programmable capacitors and/or one or more programmable resistors) may be realized by adding or modifying the passive component networks around the op-amp **809** accordingly.

(67) Moreover, it is also possible to use the proposed technique to adjust other compensator aspects. For example, by appropriately switching capacitor values and/or resistor values or gains, one can dynamically adjust the slope associated with slope compensation in a current-mode control circuit. This can be done as described for varying operating points, varying modes (e.g., buck vs. boost), or even within a switching cycle when transitioning from one mode or sub-mode to another (e.g., in moving from buck to extended buck+buck-boost mode). Such a multi-slope compensating ramp could be valuable in improving not only low-frequency dynamics over multiple modes and wide operating ranges but also ripple dynamics (i.e., as associated with “ripple instability” in current-mode-controlled converters.) One can likewise dynamically adjust PWM ramp slope in this manner, either on a long-term basis or within a switching cycle. These functions may be accomplished because the associated waveforms are generated by charging a capacitor via a current source or resistor, and the capacitor and/or resistor and/or current source can be dynamically reprogrammed as described herein.

(68) Although reference is sometimes made herein to use of the described compensators with power electronics and/or as part of a controller, it should be appreciated that compensators provided in accordance with the concepts described herein may find use in a wide variety of applications including any applications which operate with a control system. It is also recognized that compensators may be designed using electrical, mechanical, pneumatic or any other components. It should thus be appreciated that elements (including circuit elements) other than those specifically disclosed herein may be substituted where appropriate to suit the needs of a particular application, and that a person having ordinary skill in the art would understand how to select such elements and incorporate them into embodiments operating in accordance with the concepts described herein without deviating from the scope of the concepts described herein.

(69) Various embodiments of the concepts, systems, devices, structures and techniques sought to be protected are described herein with reference to the related drawings. Alternative embodiments can be devised without departing from the scope of the concepts, systems, devices, structures and techniques described herein. It is noted that various connections are set forth between elements in the above description and in the drawings. These connections, unless specified otherwise, can be direct or indirect, and the described concepts, systems, devices, structures and techniques are not intended to be limiting in this respect. Accordingly, reference to coupled elements or a coupling of elements can refer to either a direct or an indirect coupling.

(70) As an example of an indirect relationship, references in the present description to coupling element “A” to element “B” include situations in which one or more intermediate elements (e.g., element “C”) is between element “A” and element “B” as long as the relevant characteristics and functionalities of element “A” and element “B” are not substantially changed by the intermediate element(s). The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

(71) Additionally, the term “exemplary” is used herein to mean “serving as an example, instance, or illustration. Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “one or more” and “one or more” are understood to include any integer number greater than or equal to

one, i.e., one, two, three, four, etc. The terms “a plurality” are understood to include any integer number greater than or equal to two, i.e., two, three, four, five, etc. The term “connection” can include an indirect “connection” and a direct “connection.”

(72) References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

(73) For purposes of the description hereinafter, the terms “upper,” “lower,” “right,” “left,” “vertical,” “horizontal,” “top,” “bottom,” and derivatives thereof shall relate to the described structures and methods, as oriented in the drawing figures. The terms “overlying,” “atop,” “on top,” “positioned on” or “positioned atop” mean that a first element, such as a first structure, is present on a second element, such as a second structure, where intervening elements such as an interface structure can be present between the first element and the second element. The term “direct contact” means that a first element, such as a first structure, and a second element, such as a second structure, are connected without any intermediary elements.

(74) Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term) to distinguish the claim elements.

(75) The terms “approximately” and “about” may be used to mean within  $\pm 20\%$  of a target value in some embodiments, within  $\pm 10\%$  of a target value in some embodiments, within  $\pm 5\%$  of a target value in some embodiments, and yet within  $\pm 2\%$  of a target value in some embodiments. The terms “approximately” and “about” may include the target value. The term “substantially equal” may be used to refer to values that are within  $\pm 20\%$  of one another in some embodiments, within  $\pm 10\%$  of one another in some embodiments, within  $\pm 5\%$  of one another in some embodiments, and yet within  $\pm 2\%$  of one another in some embodiments.

(76) The term “substantially” may be used to refer to values that are within  $\pm 20\%$  of a comparative measure in some embodiments, within  $\pm 10\%$  in some embodiments, within  $\pm 5\%$  in some embodiments, and yet within  $\pm 2\%$  in some embodiments. For example, a first direction that is “substantially” perpendicular to a second direction may refer to a first direction that is within  $\pm 20\%$  of making a  $90^\circ$  angle with the second direction in some embodiments, within  $\pm 10\%$  of making a  $90^\circ$  angle with the second direction in some embodiments, within  $\pm 5\%$  of making a  $90^\circ$  angle with the second direction in some embodiments, and yet within  $\pm 2\%$  of making a  $90^\circ$  angle with the second direction in some embodiments.

(77) It is to be understood that the disclosed subject matter is not limited in its application to the details of construction and to the arrangements of the components set forth in the description or illustrated in the drawings. The disclosed subject matter is capable of other embodiments and of being practiced and carried out in various ways. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. As such, those skilled in the art will appreciate that the conception, upon which this disclosure is based, may readily be utilized as a basis for the designing of other structures, methods, and systems for carrying out the several purposes of the disclosed subject matter. Therefore, the claims should be regarded as including such equivalent constructions insofar as they do not depart from the spirit and scope of the disclosed subject matter.

(78) Although the disclosed subject matter has been described and illustrated in the foregoing

exemplary embodiments, it is understood that the present disclosure has been made only by way of example, and that numerous changes in the details of implementation of the disclosed subject matter may be made without departing from the spirit and scope of the disclosed subject matter.

## Claims

1. A circuit comprising: a first capacitor configured to be charged to a voltage representing state information of a compensator; a second capacitor; a buffer; and a switching network configured to transition between a first state and a second state, wherein when the switching network is in the first state, the second capacitor is connected to the output of the buffer and charged to the voltage across the first capacitor, and wherein when the switching network is in the second state, the buffer is disconnected from the second capacitor and the first capacitor and the second capacitor are connected in parallel.
2. The circuit of claim 1, wherein when the switching network is in the first state, dynamics of the compensator are determined at least in part by the first capacitor, and when the switching network is in the second state, the dynamics of the compensator are determined at least in part by a total capacitance provided by the first capacitor and the second capacitor connected in parallel.
3. The circuit of claim 2, wherein switching between the first state and the second state of the switching network affects a capacitance of the circuit, which affects a pole location of the compensator.
4. The circuit of claim 2, wherein switching between the first state and the second state of the switching network affects a capacitance of the circuit, which affects a zero location of the compensator.
5. The circuit of claim 1, wherein transitioning the switching network between the first and second states does not cause a transient impact on the voltage across the first capacitor.
6. The circuit of claim 1, wherein transitioning the switching network between the first and second states does not cause a transient impact on a voltage provided at an output of the compensator.
7. The circuit of claim 6, wherein transitioning the switching network between the first and second states does not disturb the voltage provided at the output of the compensator.
8. The circuit of claim 1, wherein the switching network comprises a complementary switch pair configured to switch between the first state and the second state.
9. The circuit of claim 8, wherein when the complementary switch pair is in the first state, a first switch is closed and a second switch is open, and when the complementary switch pair is in the second state, the first switch is open and the second switch is closed.
10. The circuit of claim 1, further comprising: a third capacitor, wherein the buffer is configured to charge the third capacitor to a voltage substantially equal to the voltage across the first capacitor, and wherein the switching network is configured to transition between at least two additional states to cause the third capacitor to switch between being charged to the voltage across the first capacitor, and being disconnected from the buffer and connected in parallel with one or both of the first capacitor and the second capacitor.
11. The circuit of claim 1, further comprising a resistor connected to one or both of the first capacitor and the second capacitor.
12. The circuit of claim 1, wherein the circuit is part of a power converter.
13. The circuit of claim 1, wherein the switching network is controlled by a control signal.
14. The circuit of claim 13, wherein the compensator is part of a system having a power converter, wherein the control signal automatically causes switches of the switching network to switch based on one or both of an operating mode and an operating point of the power converter.
15. The circuit of claim 13, wherein the control signal automatically causes switches of the switching network to switch based on an operating mode of a power converter, wherein the operating mode comprises at least one of a buck, boost, or buck+boost mode.

16. The circuit of claim 13, wherein the control signal automatically causes switches of the switching network to switch based on an operating point of a power converter, wherein the operating point comprises one or more of an input voltage, an output voltage, a conversion ratio, or a load current of the power converter.
  17. The circuit of claim 1, wherein the buffer comprises an op-amp.
  18. The circuit of claim 17, wherein the op-amp is configured to charge the second capacitor to a voltage that is related to an output voltage and a reference voltage of the compensator.
  19. A compensator circuit configured to provide an output voltage, the compensator circuit comprising: a first capacitor configured to be charged to a voltage representing state information of the compensator circuit; a second capacitor; a buffer; and a first switching network configured to transition between a first state and a second state, wherein when the first switching network is in the first state, the second capacitor is connected to the output of the buffer and charged to the voltage across the first capacitor, and wherein when the first switching network is in the second state, the buffer circuit is disconnected from the second capacitor and the first capacitor and the second capacitor are connected in parallel; a plurality of resistors; a second switching network coupled to at least one of the plurality of resistors, wherein the second switching network is configured to transition between states to cause selected ones of the plurality of resistors to be connected in parallel with each other; and a third capacitor coupled to the selected ones of the plurality of resistors connected in parallel with each other.
  20. The compensator circuit of claim 19, wherein dynamics of the compensator circuit are determined at least in part by a total resistance provided by the selected ones of the plurality of resistors connected in parallel with each other.
  21. The compensator circuit of claim 20, wherein the total resistance affects a zero location of the compensator circuit.
  22. The compensator circuit of claim 19, wherein the compensator circuit is part of a power converter.
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